

Original Research

A multi-feature deep learning system to enhance glaucoma severity diagnosis with high accuracy and fast speed

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ARTICLE INFO

Keywords:

Multi-feature
Glaucoma
Deep learning
Artificial intelligence
Fundus photograph
Visual field

ABSTRACT

Glaucoma is the leading cause of irreversible blindness, and the early detection and timely treatment are essential for glaucoma management. However, due to the interindividual variability in the characteristics of glaucoma onset, a single feature is not yet sufficient for monitoring glaucoma progression in isolation. There is an urgent need to develop more comprehensive diagnostic methods with higher accuracy. In this study, we proposed a multi-feature deep learning (MFDL) system based on intraocular pressure (IOP), color fundus photograph (CFP) and visual field (VF) to classify the glaucoma into four severity levels. We designed a three-phase framework for glaucoma severity diagnosis from coarse to fine, which contains screening, detection and classification. We trained it on 6,131 samples from 3,324 patients and tested it on independent 240 samples from 185 patients. Our results show that MFDL achieved a higher accuracy of 0.842 (95 % CI, 0.795–0.888) than the direct four classification deep learning (DFC-DL, accuracy of 0.513 [0.449–0.576]), CFP-based single-feature deep learning (CFP-DL, accuracy of 0.483 [0.420–0.547]) and VF-based single-feature deep learning (VF-DL, accuracy of 0.725 [0.668–0.782]). Its performance was statistically significantly superior to that of 8 juniors. It also outperformed 3 seniors and 1 expert, and was comparable with 2 glaucoma experts (0.842 vs 0.854, $p = 0.663$; 0.842 vs 0.858, $p = 0.580$). With the assistance of MFDL, junior ophthalmologists achieved statistically significantly higher accuracy performance, with the increased accuracy ranged from 7.50 % to 17.9 %, and that of seniors and experts were 6.30 % to 7.50 % and 5.40 % to 7.50 %. The mean diagnosis time per patient of MFDL was 5.96 s. The proposed model can potentially assist ophthalmologists in efficient and accurate glaucoma diagnosis that could aid the clinical management of glaucoma.

Abbreviations: MFDL, multi-feature deep learning; IOP, intraocular pressure; CFP, color fundus photograph; VF, visual field; CI, confidence interval; DL, deep learning; ONH, optic nerve head; GON, glaucoma optic neuropathy; OCT, optical coherence tomography; SD OCT, spectral domain optical coherence tomography; RNFL, retinal nerve fiber layer; CDR, cup disc ratio; MD, mean deviation; PSD, pattern standard deviation; DFC-DL, direct four classification deep learning; ZJU, Zhejiang University; PKU, Peking University; ROI, region of interest; CLAHE, contrast limited adaptive histogram equalization; Acc, accuracy; DD, disc diameter; SD, standard deviation; SAP, standard automated perimetry; TD, total deviation; Avg Pool, average-pooling; Conv, convolution; DS, down sampling; FC, fully connected; ID, identity; Max Pool, max-pooling; ReLU, rectified linear unit; BN, batch normalization; ROC, receiver operating characteristic; AUC, area under the ROC curve; FPR, false-positive rate; TPR, true-positive rate; AI, artificial intelligence; PGA, prostaglandin analogue.

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<https://doi.org/10.1016/j.jbi.2022.104233>

Received 29 May 2022; Received in revised form 21 September 2022; Accepted 17 October 2022

Available online 21 October 2022

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1. Introduction

Glaucoma is the main cause of irreversible blindness, affecting ~ 64.3 million patients worldwide [1–5]. It is characterized by retinal ganglion cell death, variably raised intraocular pressure (IOP), structural changes in the optic nerve head (ONH) and loss of visual field (VF) [6]. Most of the irreversible vision loss resulting from glaucoma can be prevented through early detection and preventative treatment. In clinical practice, IOP measurement accompanied by several ocular medical imaging measurements are widely used to detect glaucoma optic neuropathy (GON), color fundus photograph (CFP) and optical coherence tomography (OCT) for structure assessment and VF test for functional assessment [7]. However, manual assessment of glaucoma significantly impacts the efficiency of glaucoma screening due to labor intensive and highly dependent on image interpretation by trained specialists. As a result, the glaucoma screening approaches are not well used widely [8–10].

Recently, deep learning (DL) has shown considerable promise in glaucoma diagnosis [11,12]. Li *et al.* developed a DL system for automated classification of GON by CFP [13]. Christopher *et al.* predicted glaucomatous VF damage from spectral domain optical coherence tomography (SD OCT) data based on a DL approach [14]. Most recently, a smartphone application-based DL system was presented for glaucoma detection with VF [15]. However, single-feature information is not reliable for accurate diagnosis of glaucoma due to the possible disagreement between structural and functional measurements in glaucoma patients [16–21]. Therefore, it is of great value to combine information from multiple sources to improve the performance of diagnosis. Multi-feature learning is a hot topic in the deep learning. It refers to use data from different sensors observing a common phenomena in a complementary manner toward learning a complex task [22,23]. A back propagation network was used to discriminate between normal and glaucomatous eyes based on the combination of nine VF indices and structural data, and the diagnostic precision was improved by 4–8 % compared with separated data [24]. Other methods, such as combining structure–function indexes from OCT and VF [25,26], combining OCT with CFP for glaucoma detection [27,28], using paired VF and OCT data in FusionNet for detecting GON [29] and combining retinal nerve fiber layer (RNFL), cup disc ratio (CDR), mean deviation (MD) and pattern standard deviation (PSD) values to differentiate normal and glaucomatous eyes [30] were proposed with improved performance than the counterparts which use single-feature data. These methods suggest that multi-feature DL models are valuable.

As an irreversible but preventable disease, glaucoma requires a grading management strategy according to the disease severity. For example, patients with moderate glaucoma are recommended with IOP lowering medications while surgery may sometimes be offered as a first-line therapy for patients with severe glaucoma to lower IOP [31]. However, most existing methods only diagnose whether the patient has glaucoma, which can't meet clinical needs. To achieve individualized health care for glaucoma patients, it is essential to focus on severity classification of the glaucoma.

To address the problems outlined above, we proposed a multi-feature deep learning (MFDL) system based on IOP-CFP-VF triplet to classify the glaucoma into four severity levels. We designed a three-phase framework for glaucoma severity diagnosis from coarse to fine, which contains screening, detection and classification. Our method achieved better performance than traditional DL methods and was validated its auxiliary diagnosis ability for ophthalmologists in the clinic. As an efficient glaucoma diagnosis aided tool, it can improve the diagnostic level and efficiency of ophthalmologists, facilitate effective grading management, especially in remote and underserved communities.

2. Statement of significance

Problem or Issue	What is Already Known	What this Paper Adds
Clinical need for an automated method to diagnose glaucoma severity with high accuracy	Due to the interindividual variability in the characteristics of glaucoma onset, a single feature is not yet sufficient for monitoring glaucoma progression in isolation. And existing methods mainly focus on distinguishing health and glaucoma without further identification of the severity of glaucoma, leading to very limited significance for clinical treatment guidance.	We proposed a multi-feature deep learning system based on IOP-CFP-VF triplet to classify the glaucoma into four severity levels. A coarse-to-fine three-phase framework for glaucoma severity diagnosis has been shown to achieve better performance than traditional DL methods and was validated its auxiliary diagnosis ability for ophthalmologists in the clinic.

3. Materials and methods

3.1. Study design

Glaucoma is characterized by variably raised IOP, structural changes in the ONH and loss of VF. Therefore, the IOP-CFP-VF triplet was used to predict the glaucoma severity comprehensively in our study. However, traditional method of training one DL model that directly classifies into four classes can't satisfy precise diagnosis. Since clinical studies have proven that each feature varies with glaucoma progression to varying degrees [20,32]. In this study, we proposed a three-phase framework for glaucoma severity diagnosis from coarse to fine. It contains phase of screening, detection and classification. First, patients were screened out with IOP value. The patients with IOP greater than 21 mmHg were diagnosed as glaucoma directly [33,34]. However, it has been demonstrated that nearly 40 % of patients with otherwise characteristic glaucoma may not have elevated IOP measurements [35]. And some patients may take low IOP medications before follow-up in clinical, resulting in normal IOP, such as the moderate glaucoma in Fig. 1E. Therefore, DetectionNet was used to realize a more accurate glaucoma detection by combining CFP and VF in the second phase. Because the structural measurement exhibits the “floor effect” with glaucoma progression, which means that little structural change occurs as the glaucoma continues to deteriorate. Therefore, ClassificationNet based on VF was used for classifying the glaucoma into mild, moderate and severe in the last phase. This approach allows each feature to be used to its greatest advantage at each phase. And the result shows that it had a better performance than the direct four classification deep learning (DFC-DL) in Fig. S1. Our results also demonstrate the superiority of MFDL over the state-of-the-art methods in glaucoma severity diagnosis (Fig. S2 and Table S1).

The workflow of our study is shown in Fig. 1. First, we collected 6,371 samples from 3,509 patients in Zhejiang University (ZJU) and Peking University (PKU) for training, validating, and testing the MFDL system (Fig. 1A–B). Then we preprocessed CFP by cropping the region of interest (ROI) and contrast limited adaptive histogram equalization (CLAHE) method to emphasize features, and converted VF data into Voronoi images (Fig. 1C–E). Third, we constructed the MFDL consisting of screening, detection and classification (Fig. 1F). To demonstrate the superiority of the MFDL, we compared it with the VF-based single-feature DL (VF-DL) and CFP-based single-feature DL (CFP-DL) (Fig. 1G–H). Finally, the MFDL was used to assist 14 ophthalmologists of three levels (8 juniors, 3 seniors and 3 experts) in glaucoma severity diagnosis to improve the accuracy and speed (Fig. 1I–J).

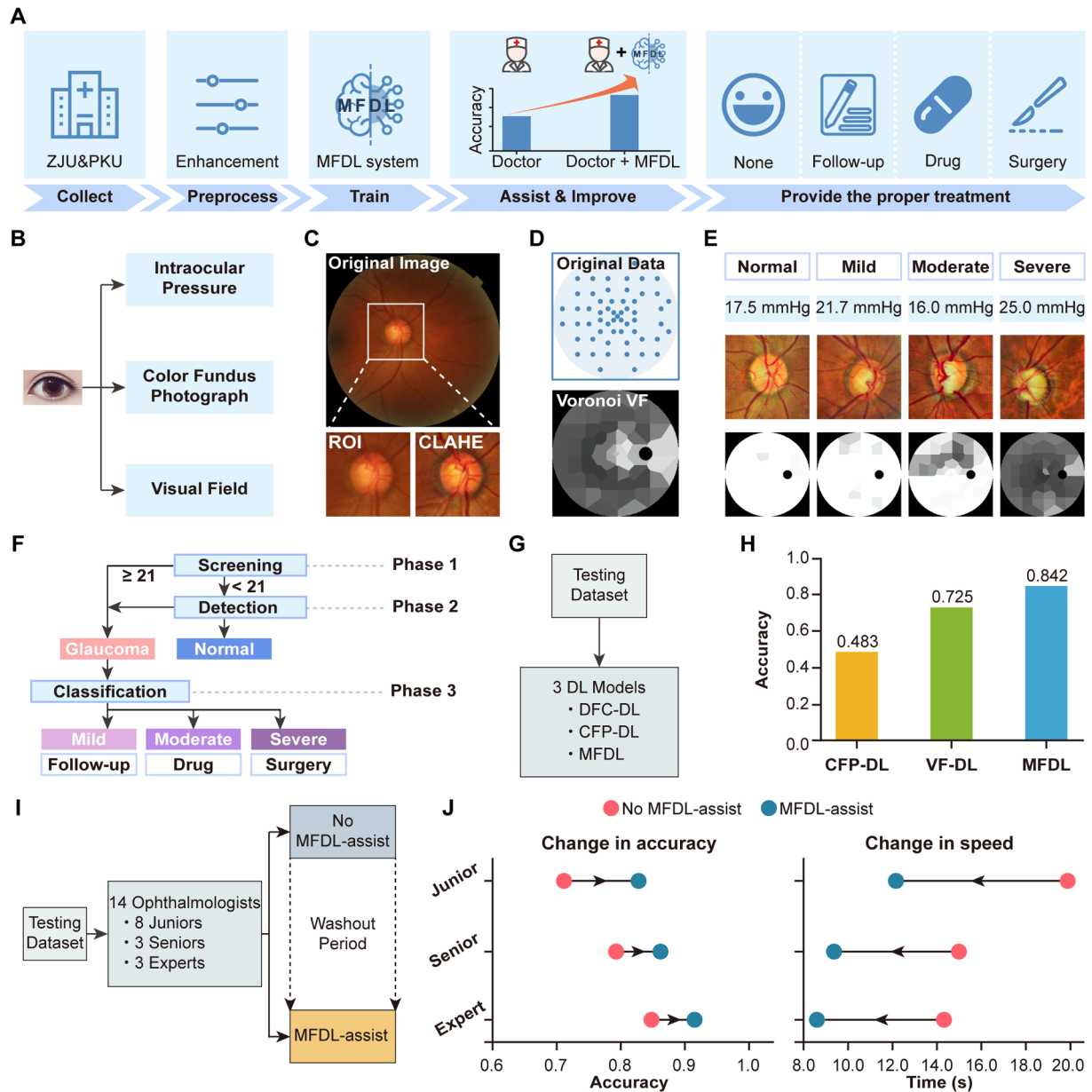


Fig. 1. Workflow of the study. (A) Workflow of MFDL system. (B) The IOP-CFP-VF triplet collected from ZJU and PKU. (C) Preprocessing of CFP by cropping ROI and CLAHE method. (D) Preprocessing of VF data. (E) Multi-feature data of four classes after preprocessing. (F) Three-phase framework of MFDL: screening, detection and classification. (G-H) Comparison between MFDL and single-feature DL. (G) Experiment design. (H) The MFDL outperformed single-feature DL. (I-J) The auxiliary diagnosis ability of MFDL in clinical. (I) Experiment design. (J) The accuracy and speed of ophthalmologists were effectively improved with the assistance of MFDL. (ZJU, Zhejiang University; PKU, Peking University; ROI, region of interest; CLAHE, contrast limited adaptive histogram equalization; DL, deep learning).

3.2. Dataset collection and labeling

This retrospective study collected ocular data from the Eye Center at the Second Affiliated Hospital of Zhejiang University School of Medicine (Zhejiang, China) and the Eye Center of Peking University Third Hospital (Beijing, China). Multi-feature data consists of the IOP, CFP and VF, which were obtained by non-contact tonometer NT-530P (NIDEK, Japan), CR-2 PLUS AF Digital Non-Mydriatic Retinal Camera (CANON, Japan) and G1 program test pattern with stimulus size III by the OCTOPUS 900 perimeter (HAAG-STREIT, Switzerland), respectively. The study was approved by the hospital's Institutional Review Board and informed consent was obtained from patients and normal controls. And it adhered to the tenets of the Declaration of Helsinki and the Health Insurance Portability and Accountability Act. We excluded cases that meet the following criteria: (1) For the VF, the false-negative rate $\geq 30\%$

and false-positive rate $\geq 30\%$; (2) For the CFP, the poor-quality images (vessels within 1 disc diameter (DD) of the optic disc margin cannot be identified, 50 % of the area is obscured and only part of optic disc is visible in the image); (3) Patients diagnosed with other retinal or optic nerve diseases.

The patients in dataset were labeled into four severity groups (normal, mild, moderate and severe) by three experienced glaucoma experts according to the American Academy of Ophthalmology guideline [16,36]. The details of the labeling criteria of glaucoma are shown in table S2. If there was divergence, a second round with communication would be applied, until a consistent grading outcome was achieved, which was considered as the conclusive grading for a given patient. Samples without consensus were excluded from the dataset. The final dataset contained 6,371 samples from 3,509 patients. Table 1 lists detailed characteristics of the patients.

Table 1
Patient characteristics in training and testing datasets.

Dataset	Classification	Samples	Eyes	Individuals	Mean age (SD)	Female	Male
Training dataset	Normal	1,635	1,537	922	27.4 (24.1)	523 (56.7 %)	399 (43.3 %)
	Mild	1,141	930	663	46.9 (18.6)	328 (49.5 %)	335 (50.5 %)
	Moderate	1,547	1,157	827	48.7 (19.4)	411 (49.7 %)	416 (50.3 %)
	Severe	1,808	1,252	912	57.2 (16.5)	412 (45.2 %)	500 (54.8 %)
Testing dataset	Normal	60	48	42	40.9 (17.4)	24 (42.9 %)	18 (57.1 %)
	Mild	60	53	45	43.5 (15.1)	18 (60.0 %)	27 (40.0 %)
	Moderate	60	54	46	49.6 (16.6)	21 (54.3 %)	25 (45.7 %)
	Severe	60	54	52	51.8 (15.1)	29 (44.2 %)	23 (55.8 %)

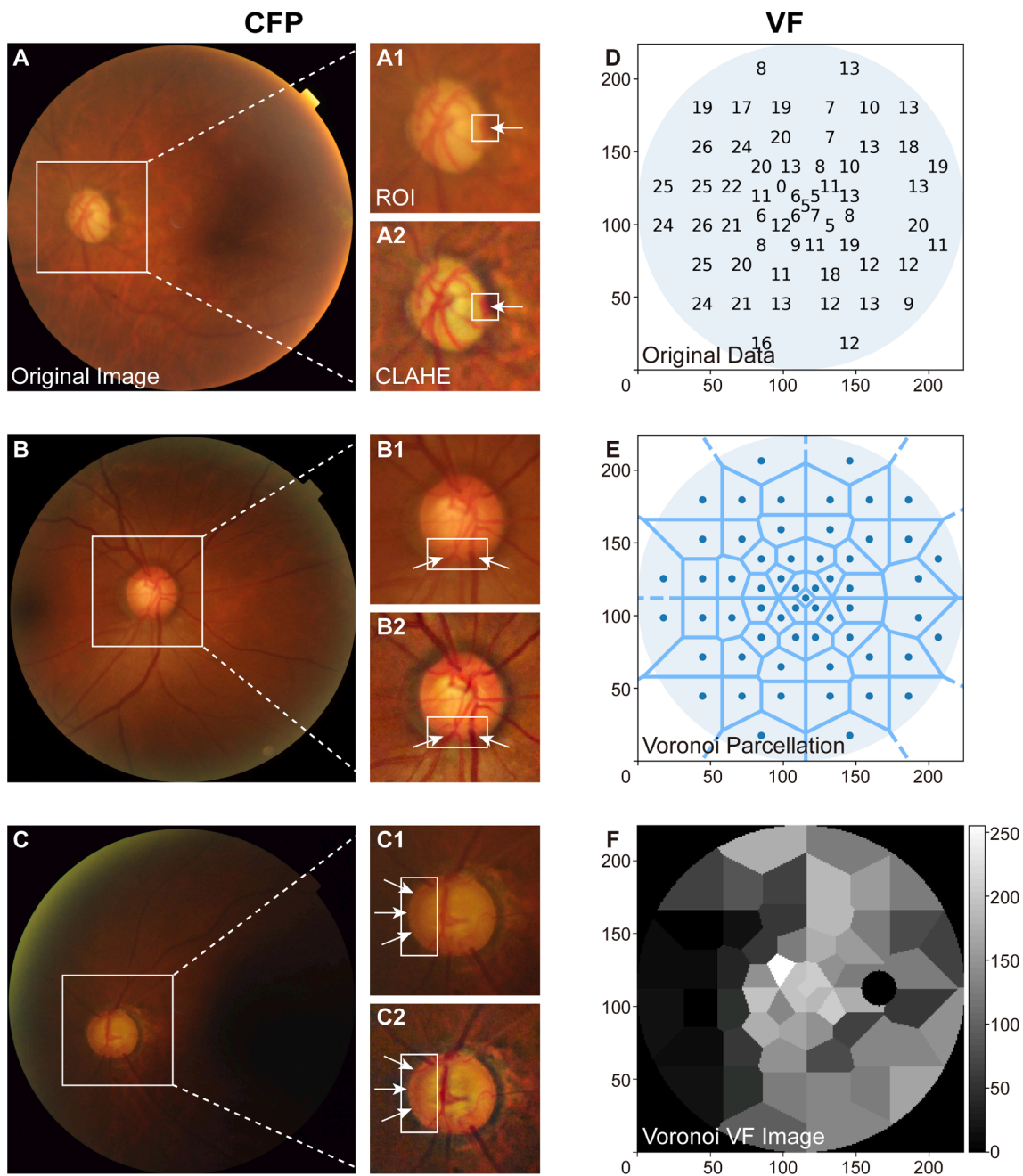
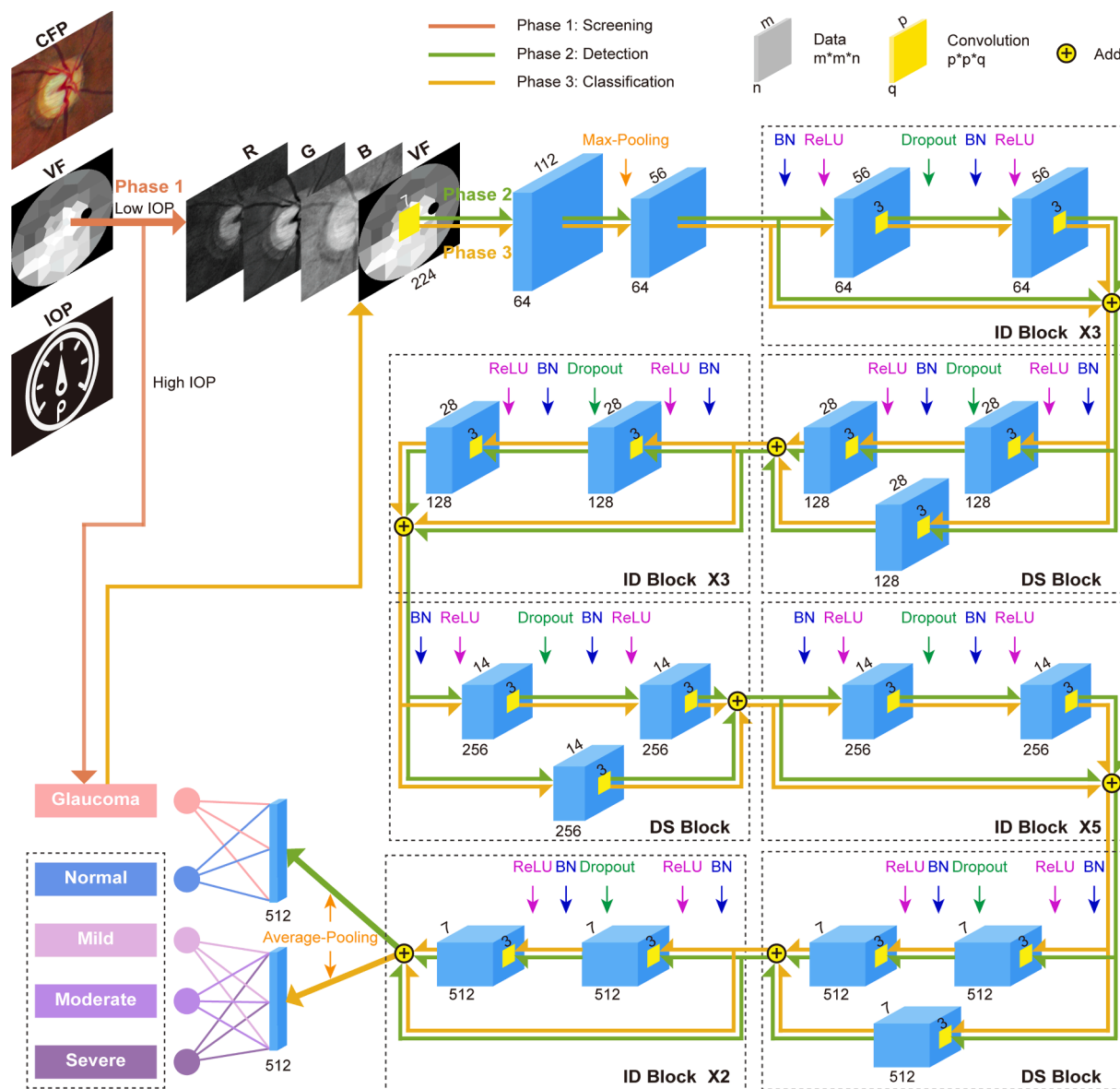


Fig. 2. Preprocessing of CFP and VF. (A-C) Three representative CFP preprocessing results. (A1, B1, C1) are the enlarged image of ONH areas cropped from the original CFP, corresponding to the white box areas in A, B, C. (A2, B2, C2) are the results of using the CLAHE method for A1, B1, and C1, respectively. The areas marked by white boxes and arrows indicate the features that have significantly improved by CLAHE: bleeding points (A1, A2), curved blood vessels (B1, B2) and cup-to-disk ratio (C1, C2). (D-F) Voronoi preprocessing of VF. (D) 59 original TD values from the VF report. (E) Voronoi parcellation. (F) Voronoi VF image. The dark colors indicate high defect and light colors indicate low defect values. Black circle depicts the blind spot. (ONH: optic nerve head; ROI: region of interest; CLAHE: contrast limited adaptive histogram equalization; TD: total deviation).

- (1) Diffuse or focal narrowing, or notching, of the optic disc rim, especially at the inferior or superior poles.
- (2) Progressive narrowing of the neuro retinal rim with an associated increase in cupping of the optic disc.
- (3) Diffuse or localized abnormalities of the parapapillary RNFL, especially at the inferior or superior poles.
- (4) Disc rim, parapapillary RNFL, or lamina cribrosa hemorrhages.
- (5) Optic disc neural rim asymmetry of the two eyes consistent with loss of neural tissue.
- (6) Large extent of parapapillary atrophy.

The CFP and VF were preprocessed before input to MFDL, as shown in Fig. 2. First, we cropped the ONH area in original CFP which is relative to glaucomatous structural impairment. Then, as one of the best enhancement methods for improving the medical imagery, CLAHE

We converted discrete VF data into Voronoi VF images to be suitable as the input of DetectionNet and ClassificationNet (Fig. 2D-F) [41,42]. Fig. 2D shows 59 original total deviation (TD) values from the VF report. The 224×224 matrix was partitioned according to the location of TD values (Fig. 2E). Each value was assigned by its closest TD value in the matrix and was normalized to 0–255 (Fig. 2F). The Voronoi VF images were augmented by applying vertical flip. All CFP and Voronoi VF left eye images were flipped to the right to maintain the same orientation.



3.4. MFDL system architecture

The architecture of MFDL is shown in Fig. 3. The IOP-CFP-VF triplet was processed by three-phase framework of screening, detection and classification to predict the severity of glaucoma. ResNet was used as the neural network in detection and classification phase, which is widely used in medical image recognition with demonstrated efficiency [43,44]. Table S3 shows the detailed architecture of it. CFP and VF images were fused as the input of the DetectionNet. The input size is set to $224 \times 224 \times 4$: the first three channels correspond to RGB channels of CFP, and the fourth channel is Voronoi VF image. The DetectionNet outputs the prediction of normal or glaucoma. In the ClassificationNet, Voronoi VF image is input and output is prediction among mild, moderate and severe glaucoma. ResNet uses 3×3 convolutional kernels to capture the local features of input, and the last fully connected layer gives the classification according to the global feature connected from all local features. The residual structure in identity and down sampling blocks allows the network to directly skip the middle convolutional layers, greatly improving the network depth and training efficiency, and reducing the problem of training stagnation caused by the vanishing gradient. The training process was accelerated by using z-score normalization and batch normalization layers. Dropout layers and L2 regularization were used to prevent overfitting. All training and testing processes were performed in TensorFlow on a NVIDIA GeForce GTX 1050 Ti GPU with Adam optimizer [45]. Loss was calculated according to the cross entropy. In the DetectionNet, the learning rate was 5×10^{-5} , batch size was 16, epoch was 150, and dropout rate was 0.2, respectively. In the ClassificationNet, we used a batch size of 16, epoch of 100, and dropout rate of 1.0. Learning rate was initialized at 1×10^{-3} , and reduced by a factor of 10 every 20 epochs.

3.5. Statistical analysis

We used confusion matrix and accuracy to evaluate the diagnosis performance. A receiver operating characteristic (ROC) curve is also an evaluation metric with its X and Y axes defined as false-positive rate (FPR) and true-positive rate (TPR), respectively [46]. In addition, the area under the ROC curve (AUC) and macro average AUC are classic quantitative metrics for two- and multiple-class classification, respectively [47]. The closer the ROC curve is to the upper left corner and the higher the AUC value, the better performance of the model.

The 95 % confidence interval (CI) was used to assess the variability in the estimates for accuracy. The two-tailed paired sample t-test was applied on the accuracy to identify significant differences in performance between the MFDL and the ophthalmologists, and $p < 0.05$ was considered to be statistically significant. All statistical analyses were performed using SPSS (v26.0, IBM) or Python (v3.6, Python Software Foundation).

4. Results

4.1. Performance of DetectionNet and ClassificationNet

We trained the DetectionNet and ClassificationNet independently (Fig. 4). The training dataset was randomly divided into two parts with a ratio of 8:2, which were used to train and validate the neural networks respectively. Fig. 4A, C plot the loss and accuracy for the training and validation datasets. Training phase was completed until accuracy and loss can't further improve. Fig. 4B-D display the ROC curves of neural networks on the validation dataset. The DetectionNet achieved an accuracy of 0.855 and AUC of 0.881 for glaucoma detection. The ClassificationNet achieved an accuracy of 0.825 and AUC of 0.956 for severity classification between mild, moderate and severe glaucoma. And the

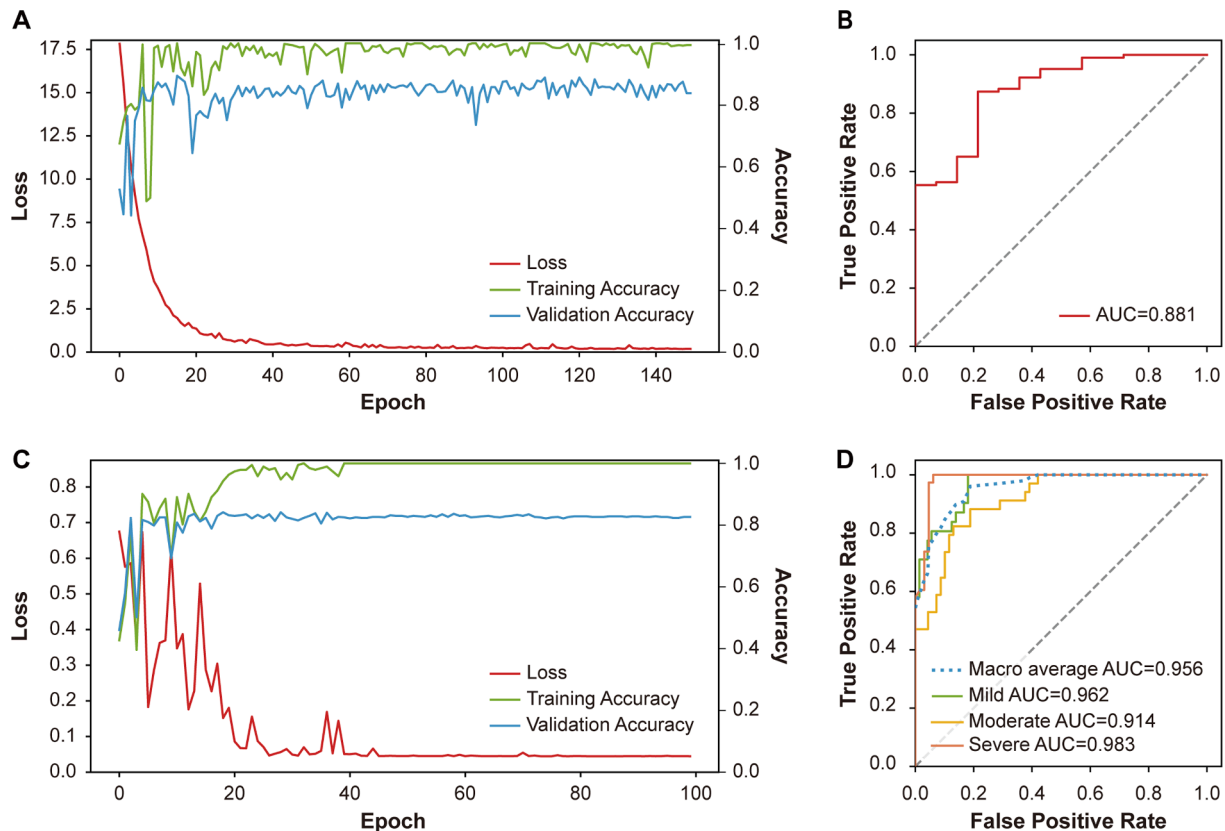


Fig. 4. Performance of DetectionNet and ClassificationNet on the training and validation datasets. (A) Training curve and (B) ROC curve of DetectionNet. (C) Training curve and (D) ROC curve of ClassificationNet. (ROC, Receiver operating characteristic).

AUC for identifying severe glaucoma was the highest at 0.983. Then the MFDL system was integrated through IOP screening, DetectionNet and ClassificationNet.

4.2. Comparison between MFDL and single-feature DL

To demonstrate the effectiveness of our multi-feature method, we compared the diagnostic performance between MFDL and single-feature DL on the testing dataset, as shown in Fig. 5. The VF-based single-feature DL (VF-DL) outperformed the CFP-based single-feature DL (CFP-DL) in moderate and severe glaucoma. But it was trapped in the detection of mild glaucoma, while CFP-DL showed better accuracy in it. It confirms that structural measures are more sensitive in early stages, whereas functional measures provide more information in advanced glaucoma. Compared with single-feature DL, MFDL achieved a higher accuracy of 0.842 (95 % CI, 0.795–0.888) than CFP-DL (accuracy of 0.483 [0.420–0.547]) and VF-DL (accuracy of 0.725 [0.668–0.782]). It achieved the most obvious enhancements in mild glaucoma (improving

accuracy about 0.150–0.284), demonstrating the advantage of MFDL in the early-stage glaucoma detection. Fig. 5C-D shows two typical cases that were misdiagnosed by the single-feature DL but were correctly diagnosed by MFDL. The left case shows the situation where the functional abnormality appears earlier than the structural abnormality. The VF shows obvious visual functional defects in the superior temporal area but the ONH is still normal in CFP. On the contrary, the right case shows that structural abnormality appears before detectable functional change. It can be clearly seen that the vertical cup-to-disk ratio in CFP is greater than 0.7 while VF tends to be normal. Both of these cases may lead to false-negative results when using a certain single-feature DL, indicating the necessity of multi-feature diagnosis combining structural and functional information.

4.3. The auxiliary diagnosis ability of MFDL in clinical

We further compared the diagnostic performance of MFDL with 14 ophthalmologists in three different levels (8 juniors, 3 seniors and 3

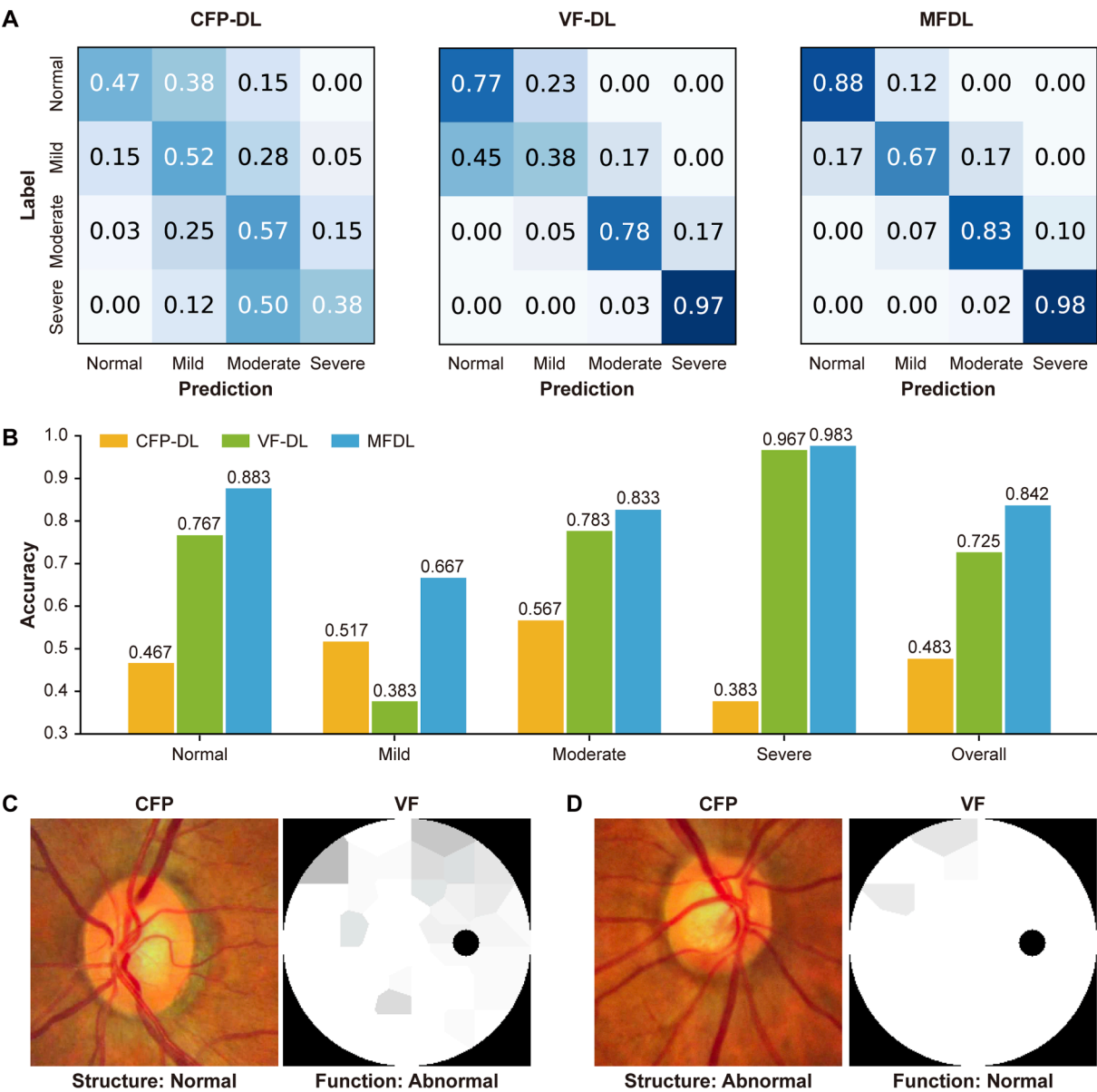


Fig. 5. Comparison between MFDL and single-feature DL on the testing dataset. (A) Normalized confusion matrix. (B) Accuracy on each class. (C-D) Two representative cases that were misdiagnosed by a certain single-feature DL but were correctly diagnosed by MFDL. (C) The case that CFP-DL misdiagnosed. (D) The case that VF-DL misdiagnosed.

experts) on the testing dataset. The expert is a clinician specializing in glaucoma for 15 years, the senior has 4–5 years of study experience, and the junior has 1–2 years of study experience, respectively. They are all from the Second Affiliated Hospital, Zhejiang University School of Medicine (Zhejiang, China) and have no prior information of the testing dataset. Their diagnostic results are shown in Fig. 6A. The confusion matrixes represent the mean of each level ophthalmologists. We compared the performance in each class between the MFDL and ophthalmologists in Fig. 6B and Table 2. The column bar shows the mean with the range of each level ophthalmologists. The MFDL achieved an accuracy of 0.842 while those of 14 ophthalmologists ranged from 0.654 to 0.858, with a mean accuracy of 0.758. Its performance was statistically significantly superior to that of 8 juniors. It also outperformed 3 seniors and 1 expert, and was comparable with 2 glaucoma experts (0.842 vs 0.854, $p = 0.663$; 0.842 vs 0.858, $p = 0.580$). The result also shows that confusing normal and mild is an important reason of misdiagnosis by ophthalmologists with lower qualifications, but MFDL performed well in this regard.

We conducted another experiment to investigate whether MFDL could help ophthalmologists improve their diagnostic performance. The experiment design is shown in Fig. 11. Each ophthalmologist was given a diagnosis probability on each patient by MFDL and was asked to make a diagnosis with the assistance of the MFDL-generated results. To avoid a

potential memorization bias, the follow-up MFDL-assisted diagnostic test was performed 10 days after the initial test. With the assistance of MFDL, juniors achieved statistically significantly higher accuracy performance, with the increased accuracy ranged from 7.50 % to 17.9 %. They were comparable to senior ophthalmologists under the help of the MFDL. Even for the senior and expert, the increased accuracy were from 6.30 % to 7.50 % and 5.40 % to 7.50 %. Performance improvements across ophthalmologists are detailed in the Fig. 6C and Table 3. The mean diagnosis time per patient of MFDL was 5.96 s, 2.1–4.4 times than that of ophthalmologists. The MFDL helped 14 ophthalmologists speed up the diagnosis, as shown in Fig. 6D. These results illustrate MFDL could effectively improve the diagnostic performance of ophthalmologists from junior to expert level participating in our test.

5. Discussion

In this study, we proposed a multi-feature deep learning (MFDL) system for glaucoma severity classification based on the IOP-CFP-VF triplet. We designed a three-phase framework containing screening, detection and classification to take the greatest advantage of each feature. The MFDL demonstrated better performance compared with the direct four classification DL and single-feature DL. Its performance was statistically significantly superior to that of juniors. It also outperformed

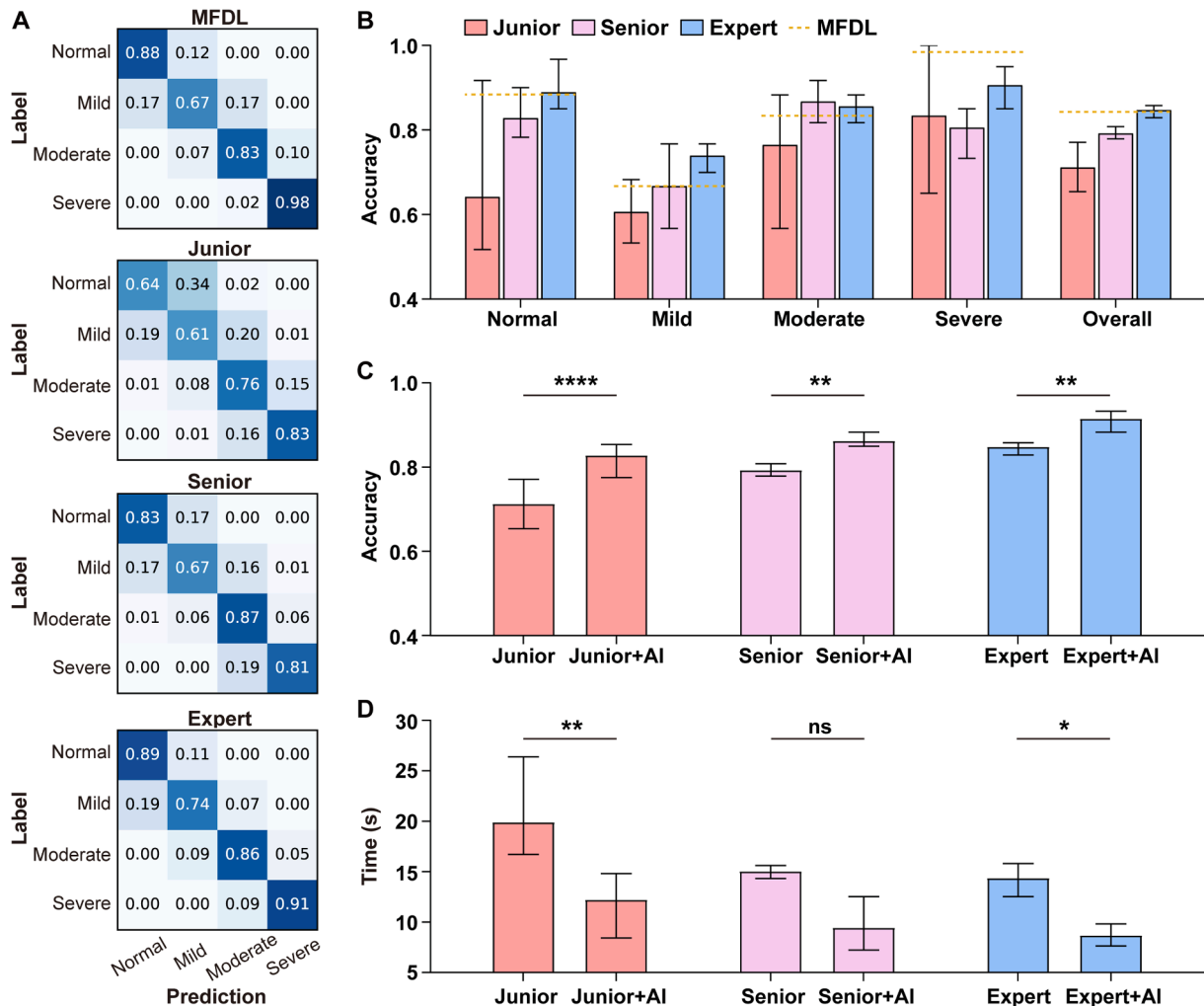


Fig. 6. Comparison between MFDL and 14 ophthalmologists on the testing dataset. (A) Normalized confusion matrix (MFDL vs the mean of each level ophthalmologists). (B) Accuracy on each class. (C-D) The auxiliary diagnosis ability of MFDL in (C) accuracy (Two-tailed paired t -test: Junior vs Junior + AI, $***p < 0.0001$; Senior vs Senior + AI, $**p = 0.0031$; Expert vs Expert + AI, $**p = 0.0092$) and (D) diagnosis time per patient (Two-tailed paired t -test: Junior vs Junior + AI, $**p = 0.001$; Senior vs Senior + AI, ns ; Expert vs Expert + AI, $*p = 0.021$). ($*p < 0.05$, $**p < 0.01$, $***p < 0.0001$, ns : non-significant difference, AI, artificial intelligence).

Table 2

Comparison between MFDL and 14 ophthalmologists on the testing dataset.

Ophthalmologist	Normal Acc	Mild Acc	Moderate Acc	Severe Acc	Overall Acc	p-value
Junior 1	0.717 (0.599–0.834)	0.617 (0.490–0.743)	0.683 (0.562–0.805)	0.650 (0.526–0.774)	0.667 (0.607–0.727)	0.000
Junior 2	0.583 (0.455–0.712)	0.600 (0.472–0.728)	0.817 (0.716–0.917)	0.817 (0.716–0.917)	0.704 (0.646–0.762)	0.000
Junior 3	0.517 (0.386–0.647)	0.550 (0.420–0.680)	0.567 (0.438–0.696)	1.000 (1.000–1.000)	0.658 (0.598–0.719)	0.000
Junior 4	0.617 (0.490–0.743)	0.600 (0.472–0.728)	0.750 (0.637–0.863)	0.650 (0.526–0.774)	0.654 (0.594–0.715)	0.000
Junior 5	0.700 (0.581–0.819)	0.533 (0.403–0.663)	0.850 (0.757–0.943)	0.850 (0.757–0.743)	0.733 (0.677–0.790)	0.001
Junior 6	0.917 (0.845–0.989)	0.583 (0.455–0.712)	0.717 (0.599–0.834)	0.867 (0.778–0.955)	0.771 (0.717–0.824)	0.019
Junior 7	0.550 (0.420–0.680)	0.683 (0.562–0.805)	0.883 (0.800–0.967)	0.867 (0.778–0.955)	0.746 (0.690–0.801)	0.003
Junior 8	0.533 (0.403–0.663)	0.683 (0.562–0.805)	0.850 (0.757–0.943)	0.967 (0.920–1.000)	0.758 (0.704–0.813)	0.006
Senior 1	0.783 (0.676–0.891)	0.667 (0.544–0.789)	0.867 (0.778–0.955)	0.833 (0.736–0.930)	0.788 (0.735–0.840)	0.052
Senior 2	0.800 (0.696–0.904)	0.767 (0.656–0.877)	0.817 (0.716–0.917)	0.850 (0.757–0.943)	0.808 (0.758–0.858)	0.311
Senior 3	0.900 (0.822–0.978)	0.567 (0.438–0.696)	0.917 (0.845–0.989)	0.733 (0.618–0.849)	0.779 (0.726–0.832)	0.059
Expert 1	0.850 (0.757–0.943)	0.767 (0.656–0.877)	0.883 (0.800–0.967)	0.917 (0.845–0.989)	0.854 (0.809–0.899)	0.663
Expert 2	0.967 (0.920–1.000)	0.750 (0.637–0.863)	0.867 (0.778–0.955)	0.850 (0.757–0.943)	0.858 (0.814–0.903)	0.580
Expert 3	0.850 (0.757–0.943)	0.700 (0.581–0.819)	0.817 (0.716–0.917)	0.950 (0.893–1.000)	0.829 (0.781–0.877)	0.687
MFDL	0.883 (0.800–0.967)	0.667 (0.544–0.789)	0.833 (0.736–0.930)	0.983 (0.950–1.000)	0.842 (0.795–0.888)	*

(Accuracy is presented with 95% CIs. Acc, accuracy. *There is no p value.).

Table 3

14 Ophthalmologists performance with and without MFDL.

Ophthalmologist	Diagnosis Acc			p-value
	No AI	With AI	With AI - No AI	
Junior 1	0.667 (0.607–0.727)	0.800 (0.749–0.851)	0.133 (0.071–0.196)	0.000
Junior 2	0.704 (0.646–0.762)	0.838 (0.790–0.885)	0.133 (0.084–0.183)	0.000
Junior 3	0.658 (0.598–0.719)	0.838 (0.790–0.885)	0.179 (0.108–0.251)	0.000
Junior 4	0.654 (0.594–0.715)	0.775 (0.722–0.828)	0.121 (0.057–0.184)	0.000
Junior 5	0.733 (0.677–0.790)	0.842 (0.795–0.888)	0.108 (0.051–0.166)	0.000
Junior 6	0.771 (0.717–0.824)	0.854 (0.809–0.899)	0.083 (0.025–0.142)	0.005
Junior 7	0.746 (0.690–0.801)	0.833 (0.786–0.881)	0.088 (0.039–0.136)	0.000
Junior 8	0.758 (0.704–0.813)	0.833 (0.786–0.881)	0.075 (0.024–0.126)	0.004
Senior 1	0.788 (0.735–0.840)	0.850 (0.805–0.896)	0.063 (0.016–0.109)	0.009
Senior 2	0.808 (0.758–0.858)	0.883 (0.842–0.924)	0.075 (0.028–0.122)	0.002
Senior 3	0.779 (0.726–0.832)	0.850 (0.805–0.896)	0.071 (0.027–0.114)	0.001
Expert 1	0.854 (0.809–0.899)	0.925 (0.891–0.959)	0.071 (0.020–0.121)	0.006
Expert 2	0.858 (0.814–0.903)	0.933 (0.902–0.965)	0.075 (0.033–0.117)	0.001
Expert 3	0.829 (0.781–0.877)	0.883 (0.842–0.924)	0.054 (0.000–0.110)	0.058

(Accuracy is presented with 95% CIs. Acc, accuracy.).

seniors and was comparable with glaucoma experts. Furthermore, MFDL could effectively improve the diagnostic performance of ophthalmologists from junior to expert level in accuracy and speed.

We demonstrated the superiority of our algorithm by comparing

MFDL with the direct four classification deep learning (DFC-DL). MFDL achieved a higher accuracy of 0.842 (95 % CI, 0.795–0.888) than the direct four classification deep learning (DFC-DL, accuracy of 0.513 [0.449–0.576]). This may because it is difficult for one neural network to learn and trade off multiple features well. And clinical studies have proven that structural measures are more sensitive in early stages, whereas functional measures provide more information in advanced disease. The results demonstrate traditional method of training one DL model that directly classifies into four classes can't satisfy precise diagnosis. MFDL allowed each feature to be used to its greatest advantage at each phase by using the three-phase framework for glaucoma severity diagnosis from coarse to fine.

We demonstrated the superiority of multi-feature data by comparing MFDL with single-feature DL. MFDL achieved a higher accuracy of 0.842 (95 % CI, 0.795–0.888) than CFP-DL (accuracy of 0.483 [0.420–0.547]) and VF-DL (accuracy of 0.725 [0.668–0.782]). Some clinical studies have proven that there is interindividual variability in the progression of glaucoma, which means some patients show structural changes prior to detectable VF abnormalities while others experience the converse [17,36], such as the cases in Fig. 5C-D. Therefore, a single test, either of structure or of function, is not yet sufficient for monitoring glaucoma progression in isolation. Our results prove that the multi-feature data provides complementary information for glaucoma severity classification, which is consistent with the practice of clinical diagnosis.

We proved the superiority of MFDL in early glaucoma detection. The diagnosis of glaucoma is frequently delayed because it may be asymptomatic until developed to a relatively late stage [6], which means the early detection is much challenging. Our results show that MFDL boosted the performance in identifying mild glaucoma by an accuracy of 0.384, 0.150 and 0.284 compared with DFC-DL, CFP-DL and VF-DL. The accurate early glaucoma detection is essential for prompt treatment to slow the disease progression.

We also conducted experiments to investigate the clinical auxiliary diagnosis ability of MFDL. Its performance was statistically significantly superior to that of juniors. It also outperformed seniors and was

comparable with glaucoma experts. With the assistance of MFDL, juniors achieved statistically significantly higher accuracy performance, with the increased accuracy ranged from 7.50 % to 17.9 %, and that of seniors and experts were 6.30 % to 7.50 % and 5.40 % to 7.50 %. The mean diagnosis time per patient of MFDL was 5.96 s. As an effective auxiliary tool for glaucoma diagnosis, MFDL is suitable for medical students and ophthalmologists with lower qualifications. It can alleviate the huge demand for diagnostic expertise when the health system is overloaded or in remote areas.

As an irreversible but preventable disease, glaucoma requires a grading management strategy according to the disease severity. However, most existing methods only diagnose whether the patient has a glaucoma. To facilitate effective grading management, MFDL classified the severity of glaucoma into normal, mild, moderate and severe glaucoma. It can provide clinicians with appropriate follow-up and treatment decisions by enhancing the auxiliary diagnosis ability of severity classification. For instance, a shorter follow-up period such as 1–2 months is recommended for patients with mild glaucoma [36]. For patients with moderate glaucoma, IOP lowering medications such as generic prostaglandin analogue (PGA) may be more appropriate [31,36]. In addition, surgery may sometimes be offered as a first-line therapy for patients with severe glaucoma [48]. Overall, MFDL is dedicated to properly allocate monitoring and treatment in glaucoma management and avoid overburdening both patient and health care systems.

There are several limitations in this study. First, a larger multi-center study is necessary to further validate the performance of MFDL. Second, some clinical researches have shown that screening accuracy is improved when OCT is added to CFP [28]. In addition, advanced age and a positive family history of glaucoma are also risk factors of glaucoma [49–51]. MFDL can be developed by integrating more information (such as OCT image, age and positive family history of glaucoma, etc.) to further improve the accuracy of glaucoma severity diagnosis. Moreover, other interesting researches can be explored for the effectiveness and generalizability of MFDL in other disease areas in the future.

6. Conclusion

In conclusion, we proposed a multi-feature deep learning (MFDL) system for glaucoma severity classification based on the IOP-CFP-VF triplet. We designed a three-phase framework which showed better performance compared with traditional methods. As an efficient auxiliary diagnostic tool, MFDL could assist ophthalmologists who are not specialized in glaucoma severity diagnosis, especially in remote and poor communities. It allows for rapid glaucoma screenings during primary care visits and provides individualized health care advices for glaucoma patients, which is a promising approach to benefit a wide patient population.

Funding

This work was supported by Key R&D Program of Zhejiang Province (2022C03034); National Natural Science Foundation of China (61975178); Natural Science Foundation of Zhejiang Province (LR20F050002); Zhejiang leading innovation and entrepreneurship team (202099144); CAMS Innovation Fund for Medical Sciences (2019-I2M-5-057); Fundamental Research Funds for the Central Universities; Alibaba Cloud.

Competing interests: Authors declare that they have no competing interests.

Data and materials availability: Data cannot be shared publicly due to the violation of patient privacy and lack of informed consent for data sharing. Data are available from the Second Affiliated Hospital of Zhejiang University, Department of Ophthalmology (contact Prof. Juan Ye, yejuan@zju.edu.cn) for researchers who meet the criteria for access to confidential data.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We are very grateful to the clinicians from the Eye Center at the Second Affiliated Hospital of Zhejiang University School of Medicine (Zhejiang, China) for participating in data collection, annotation and the human-machine comparison. We also thank the Eye Center of Peking University Third Hospital (Beijing, China) for contributions to data collection.

KS, WG and JY contributed to the conception and design of the study. YX contributed to the literature search and data extraction. YX and JZZ contributed to algorithm design. YX, JZZ and XLH contributed to data analysis and interpretation. YX contributed to writing the manuscript. All authors contributed to critical revision of the manuscript. All authors approved the manuscript.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jbi.2022.104233>.

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